

Problem

A relay has a resistance of $200\ \Omega$ and an inductance of 500 mH . The relay contacts close when the current through the coil reaches $175\ \mu\text{A}$. What time elapses between the application of 110 V to the coil and contact closure?

Step-by-step solution

Step 1 of 6

The current through the coil is,

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}} \dots\dots (1)$$

Determine the initial value of current through the inductor.

$$i(0) = 0$$

Step 2 of 6

Determine the final value of current through the inductor.

$$i(\infty) = \frac{V_s}{R}$$

Substitute 110 V for V_s and $200\ \Omega$ for R in the equation.

$$\begin{aligned} i(\infty) &= \frac{110}{200} \\ &= 0.55\text{ A} \end{aligned}$$

Step 3 of 6

Determine the value of time constant of the circuit.

$$\tau = \frac{L}{R}$$

Substitute 500 mH for L and $200\ \Omega$ for R in the equation.

$$\begin{aligned} \tau &= \frac{500(10^{-3})}{200} \\ &= 2.5\text{ ms} \end{aligned}$$

Step 4 of 6

Substitute 0.55 A for $i(\infty)$ and 0 for $i(0)$ in equation (1).

$$i(t) = 0.55 \left[1 - e^{-\frac{t}{\tau}} \right] \text{ A}$$

Substitute $175\text{ }\mu\text{A}$ for $i(t_d)$ in the equation.

$$(175)(10^{-6}) = 0.55 \left[1 - e^{-\frac{t_d}{\tau}} \right]$$

$$\begin{aligned} \left[1 - e^{-\frac{t_d}{\tau}} \right] &= \frac{(175)(10^{-6})}{0.55} \\ &= 0.0003182 \end{aligned}$$

Step 5 of 6

Simply the equation further,

$$e^{-\frac{t_d}{\tau}} = 0.99968$$

$$\begin{aligned} e^{\frac{t_d}{\tau}} &= \frac{1}{0.99968} \\ &= 1.0003183 \end{aligned}$$

Step 6 of 6

Take the natural logarithm of both sides.

$$\begin{aligned} \frac{t_d}{\tau} &= \ln(1.0003183) \\ t_d &= \tau \ln(1.0003183) \end{aligned}$$

Substitute 2.5 ms for τ in the equation.

$$\begin{aligned} t_d &= (2.5) \ln(1.0003183) \text{ ms} \\ &= 0.0007956 \text{ ms} \\ &= 0.7956 \text{ }\mu\text{s} \end{aligned}$$

Therefore, the time elapses between the application of 110 V to the coil and contact closure is

$$\boxed{0.7956\text{ }\mu\text{s}}.$$